



Installation Instruction

Installation Instruction Number: 5414689

Released Date: 19-nov-2020

Thermal Recirculation Valve Upfit Kit Installation Instructions

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Product Affected

- X15 CM2350 X114B
- X15 CM2350 X116B

X15 CM2350 X114B

X15 CM2350 X114B uses two fuel line routings, configuration A or configuration B.

- For engines equipped with configuration A fuel lines, currently there is **no** service kit available.
- For engines equipped with configuration B fuel lines, kit part number 5633513 can be used.

Configuration A

There may be an overlap in engine serial number (ESN). Verify the vehicle's appropriate fuel supply tube configuration and part number before commencing repairs. See Figure 1 and Figure 2 for visual confirmation and Table 2 for part number verification.

Approximate ESN range for configuration A: 79890940 to 80038419. For ESN outside of this range confirm part numbers from Table 2.

No service kit is available for this configuration.

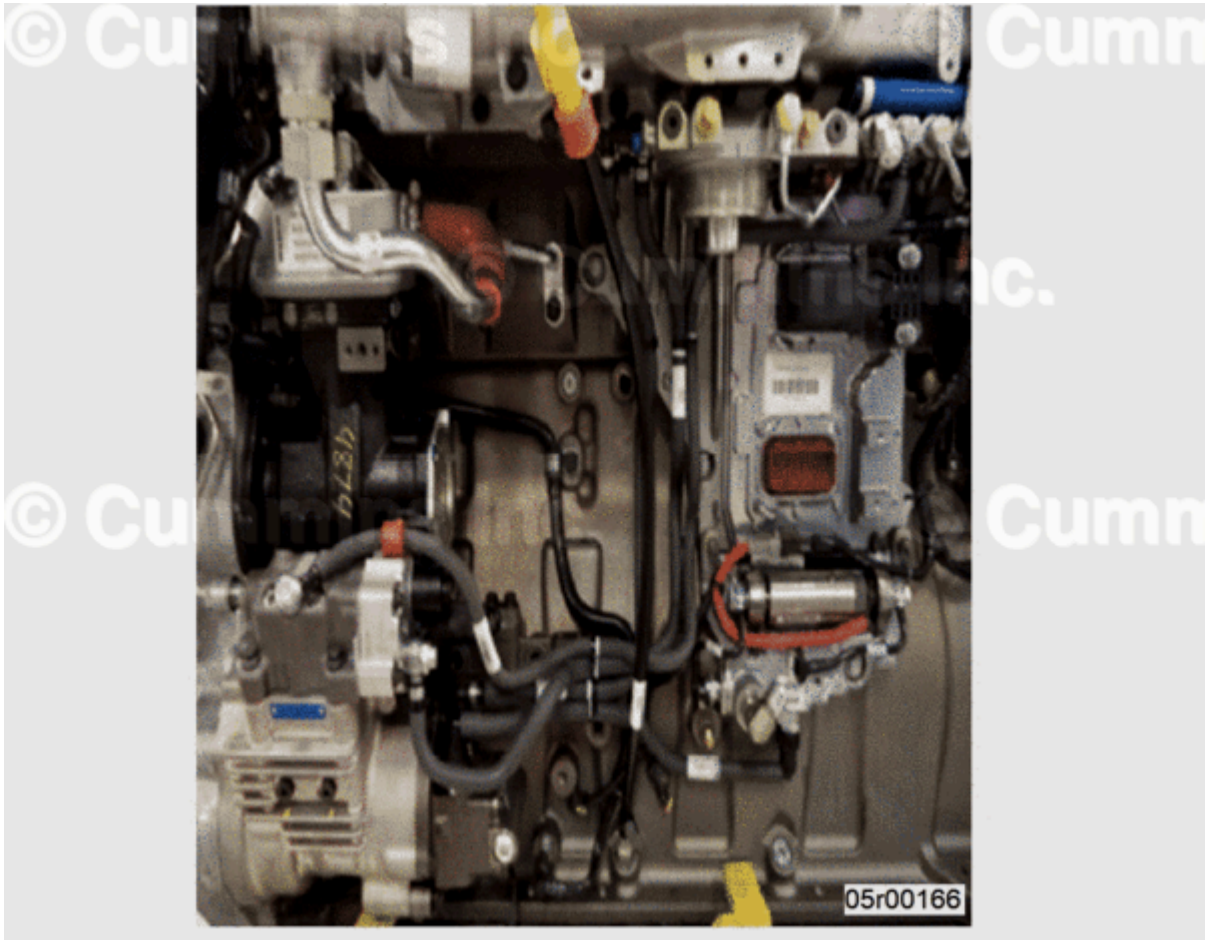


Figure 1, Representation of Configuration A.

Configuration B

There may be an overlap in engine serial number (ESN). Verify the vehicle’s appropriate fuel supply tube configuration and part number before commencing repairs. See Figure 1 and Figure 2 for visual confirmation and Table 2 for part number verification.

Approximate engine serial number (ESN) range for configuration B: 79999784 to 80276968 for ESN outside of this range confirm part numbers from Table 2.

A service kit is available, Part Number 5633513, includes components required to upfit to TRV 105°F thermostat with integrated check valve and extended fuel filter. See Table 1 for list of components

Table 1, X15 CM2350 X114B and ISX15 CM2350 X101 Configuration B Service Kit 5633513		
Part Description	Part Number	Quantity
WASHER,SEALING	3963990	4
CLAMP,WIRE TIE	4952223	1
TUBE,FUEL DRAIN	5536886	1
TUBE,FUEL DRAIN	5537254	1
MANIFOLD,FUEL	5612313	1

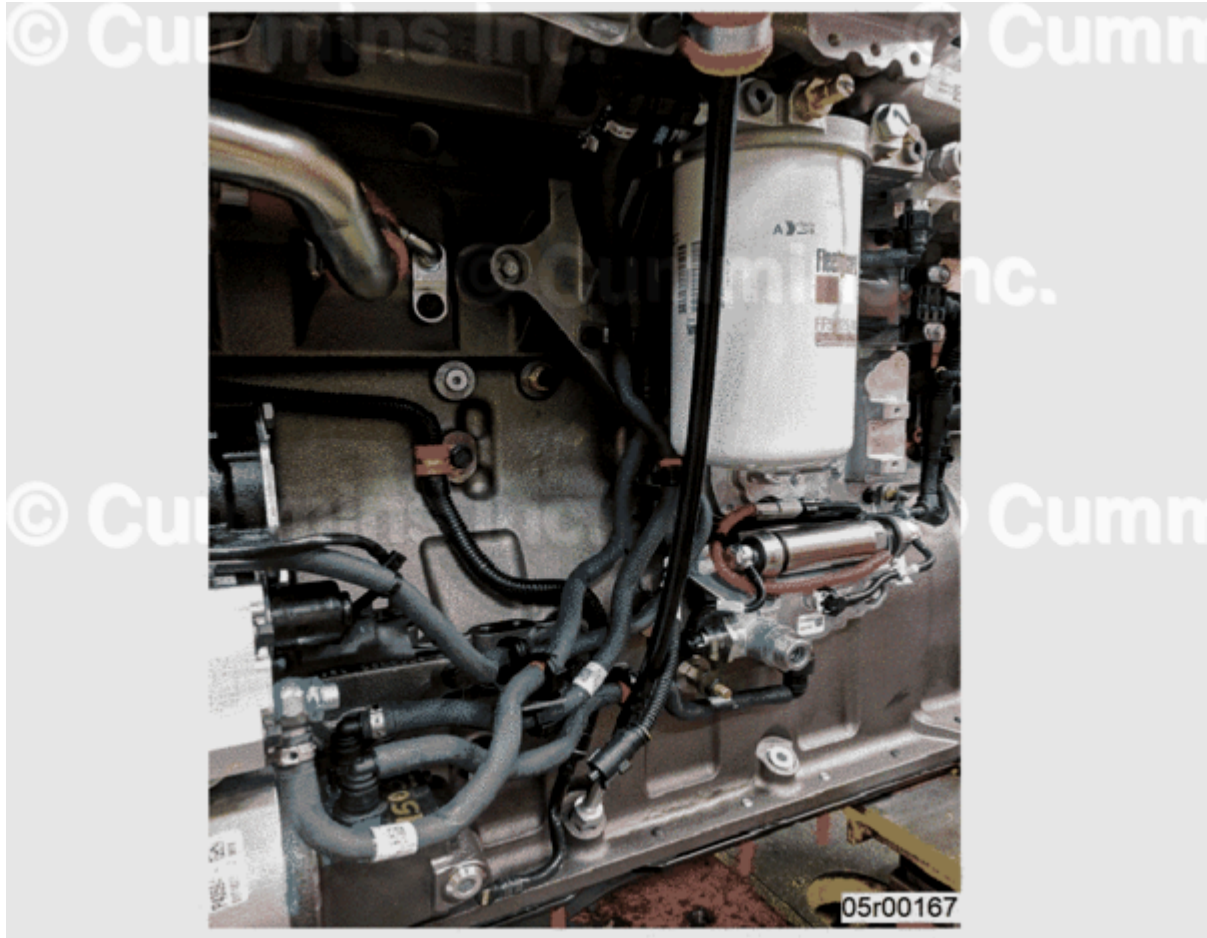


Figure 2, Representation of Configuration B.

Engines might have a nylon fuel supply line or a steel fuel supply line. See Table 2.

Table 2, Part Number Verification

Part Description	Nylon Fuel Supply Line	Steel Fuel Supply Line
Tube, Fuel Supply Configuration A	3691498	5575157
Tube, Fuel Supply Configuration B	5440474	5558764

X15 CM2350 X116B

X15 CM2350 X116B can use service kit, Part Number 5633511, to upfit to new TRV and upfit to extend fuel filter. Service kit part number 5633511, includes components required to upfit to TRV 105°F thermostat with integrated check valve and extended fuel filter. See Table 3 below for list of components.

Table 3, X15 CM2350 X116B Service Kit 5633511

Part Description	Part Number	Quantity
WASHER,SEALING	3963990	2
TUBE,FUEL DRAIN	5536886	1

Table 3, X15 CM2350 X116B Service Kit 5633511

Part Description	Part Number	Quantity
TUBE,FUEL DRAIN	5536891	1
MANIFOLD,FUEL	5612313	1

Parts Required

The parts required to install are provided in Table 1 above (X15 CM2350 X114B and ISX15 CM2350 X101) and Table 3 (X15 CM2350 X116B).

Personal Protection Equipment (PPE)

- Completing a Job Safety Assessment (JSA) prior to performing work helps identify job safety hazards and prevent incidents.
- To reduce the possibility of personal injury, personal protective equipment (PPE) must be utilized. Reference General Safety Instructions Procedure 204-006 in Section I of the corresponding Service Manual.

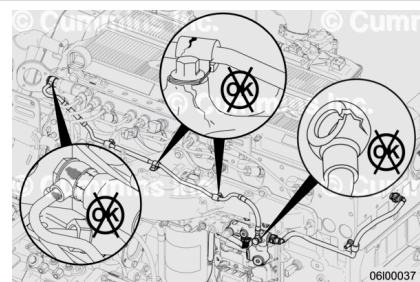
Installation Tools

No special service tools are required for installation.

Initial Check

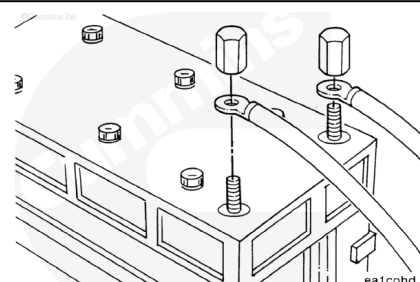
Inspect the fuel drain lines for any signs of leaks, cracks, chafing, and loose or broken brackets.

Repair or replace any damaged parts.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Disconnect the batteries. See equipment manufacturer service information.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

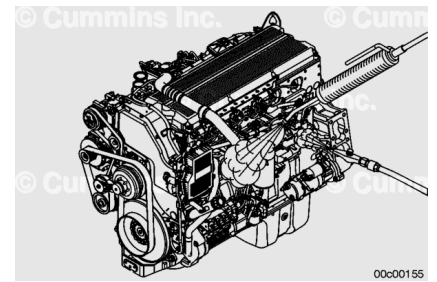
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.

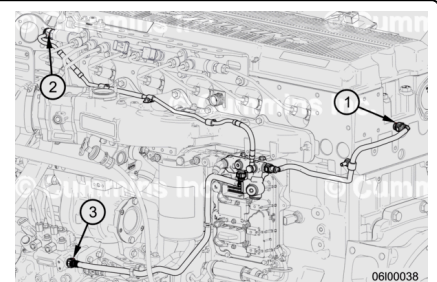
- Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.) which would expose the fuel system or internal engine component to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine.
- Clean the fuel drain line and mating components with electrical contact cleaner, Part Number 3824510, or equivalent.
- To prevent damage from debris and contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.



Remove

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



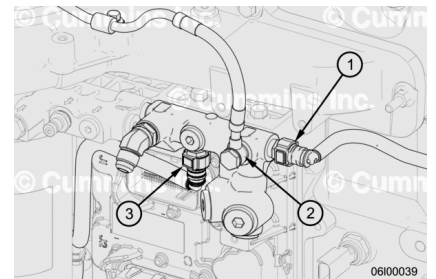
⚠ WARNING ⚠

Do not vent the fuel system on a hot engine; this can cause fuel to spill onto a hot exhaust manifold, which can cause a fire.

- Remove the injector drain line connected at the back of the cylinder head by pressing the tab on the quick disconnect (1)
- Leave the fuel rail drain line (2) connected to the fuel rail.
- Remove the fuel pump drain line (3) for efficiency series engines press the tab on the quick disconnect and remove from the fuel pump, for performance series engines remove the M12 banjo bolt connecting the drain line to the fuel pump.

⚠ CAUTION ⚠

Do not use a screwdriver or pliers on the locking tabs. This may result in damage to the quick disconnect fitting.



Disconnect the drain lines from the drain manifold

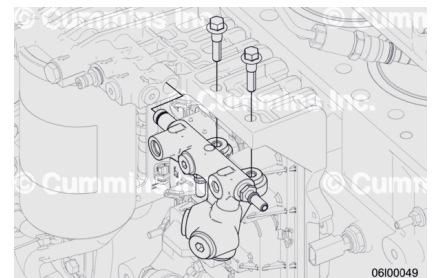
- Press the tab on the quick disconnect to remove the injector drain line (1).
- Remove the M12 banjo bolt from the fuel rail pressure relief valve (2).
- Press the tab on the quick disconnect to remove the fuel pump drain (3).

For the rail pressure relief drain line **only** remove the Banjo bolt at the Fuel Manifold (2). Do **not** remove the banjo bolt or P clips on the rail pressure relief drain line.

Remove any P-clip capscrews, studs, banjo bolts and sealing washers when removing these lines.

Remove the air intake manifold from the engine. See corresponding Service Manual. Reference Procedure 010-023 in Section 10.

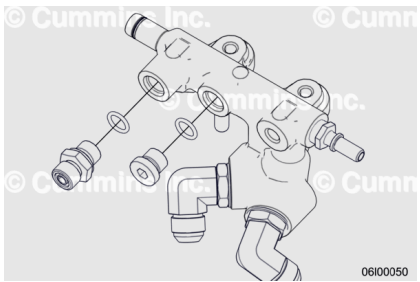
Remove the mounting capscrews and the thermal fuel recirculation valve.



Disassemble

Remove the OEM fuel return fitting.

Remove the threaded plug.



Inspect for Reuse

If OEM Fittings need replaced, see Table 3 for OEM Part numbers.

If customer needs to connect OEM tank to the TRV, additional tubing and fittings are in Table 4 below.

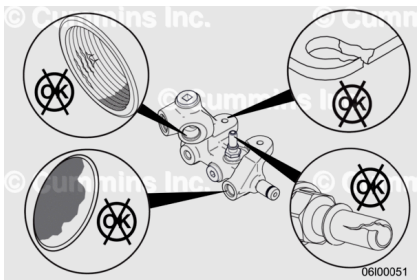


Table 3, OEM Fuel Return Fittings

OEM	TRV Fitting Information
Navistar	4062511C1
Volvo	22798467
Daimler	Fitting with M22 x 1.5 Threads
Cummins Quick Connect	3692933

Table 4, OEM Fitting to Connect fuel tank to TRV Fitting

OEM	Part Information		
Navistar	Part #	Quantity	Description
	3571312C1	2	Nylon Black 5/8 Tube
	3559213C2	3	Quick Connect
	3578152C1	1	Compression fuel outlet fitting
Daimler	#8 and #10 Hose		

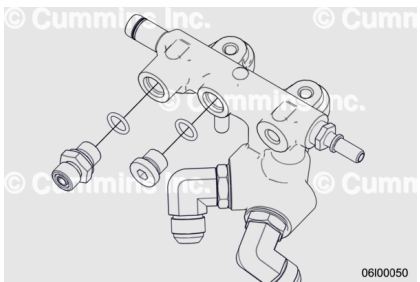
Assemble

Install the OEM fuel return fitting into new fuel drain manifold, Part number 5612313.

Torque Value: 40 n•m [30 ft-lb]

Install the threaded plug.

Torque Value: 37 n•m [27 ft-lb]



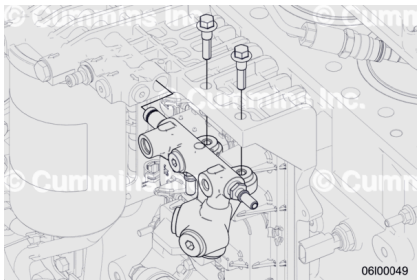
Install (If Needed)

Install the fuel drain manifold capscrews through the fuel filter head casting and into the new fuel drain manifold (Part number 5612313).

Tighten the capscrews.

Torque Value: 45 n·m [33 ft-lb]

Install the air intake manifold from the engine. See corresponding Service Manual. Reference Procedure 010-023 in Section 10.

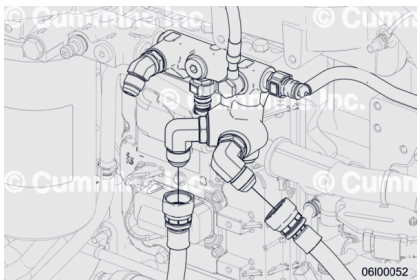


Install the fuel supply lines.

- Connect the fuel supply line at the right side M22 x 1.5 port of the thermal recirculation manifold.
- Connect the fuel supply line at the left side M22 x 1.5 port of the TRV manifold.

Torque Value: 47 n·m [35 ft-lb]

See corresponding Service Manual. Reference Procedure 200-001 in Section F.



Install the new drain lines on the engine.

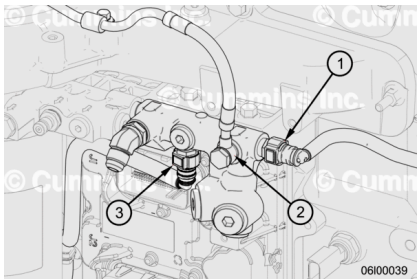


Table 5, Location of new Drain Line Installation

Identify from Picture	New Part Number	Notes
1	5536886	-
2	No new Part	Use quantity x2 new sealing washers when installing banjo bolt, Part Number 3963990.
3	5537254 (X15 CM2350 X114B and ISX15 CM2350 X101) 5536891 (X15 CM2350 X116B)	For X15 CM2350 X114B and ISX15 CM2350 X101 use two new sealing washer, Part Number 3963990 when installing fuel pump drain line.

Finishing Steps (if needed)

Loosely install the P-clip capscrews, stud, banjo fittings, and sealing washers for each drain line.

Tighten the P-clip capscrews, and banjo bolts.

Install quick disconnect fittings, if equipped.

Torque Values:

M12 Banjo Bolts and Check Valve

Torque Value: 24 n•m [212 in-lb]

Quick Disconnect Check Valve With Copper Washer

Torque Value: 54 n•m [40 ft-lb]

M16 Banjo Bolts

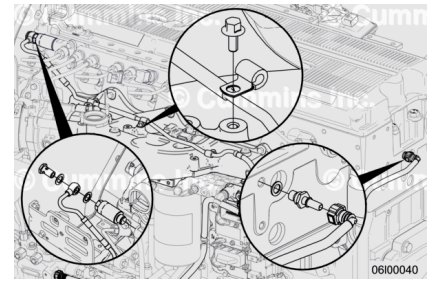
Torque Value: 40 n•m [29 ft-lb]

P-Clip Capscrew

Torque Value: 20 n•m [177 in-lb]

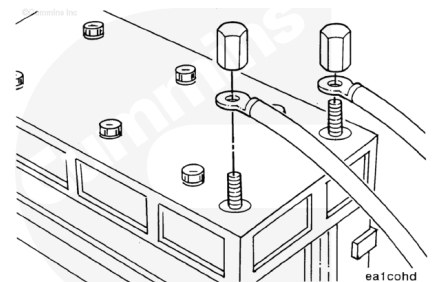
P-Clip Stud

Torque Value: 45 n•m [33 in-lb]



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Connect the fuel lines. Make sure the connection gives an audible click and then give a slight pull on the fuel line to verify the connector is fully seated and locked.
- Connect the OEM fuel return line. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Bleed air from the fuel system. See corresponding Service Manual. Reference Procedure 006-003 in Section 3.
- Operate the engine. Check for leaks.

Document History

Date	Details
2020-11-19	Module Created
2021-11-15	Addition of a missing step that aligns with the change in Service Manual.

Last Modified: 03-Jan-2022
